

Chapter 4: Atomic Structure

Book: Modern Physics by Beiser (Chapters 4, 5 and 6)

Modern Physics by Kenneth S Krane

Modern Physics by Richtmyer

Atomic Structure

- The Nuclear Atom

An atom is largely empty space

- Electron Orbits

The planetary model of the atom and why it fails

- Atomic Spectra

Each element has a characteristic line spectrum

- The Bohr Atom

Electron waves in the atom

- Energy Levels and Spectra

A photon is emitted when an electron jumps from one energy level to a lower level

- Correspondence Principle

The greater the quantum number, the closer quantum physics approaches classical physics

- Nuclear Motion

The nuclear mass affects the wavelengths of spectral lines

- Atomic Excitation

How atoms absorb and emit energy

Timeline of Atomic Theory

Greek Model
400 BC
Democristus

(Aristotle's 4 Elements)

water fire earth air

Rutherford Model
1911

Wave Model
Modern

Dalton Model
1803

atoms are indivisible
particles

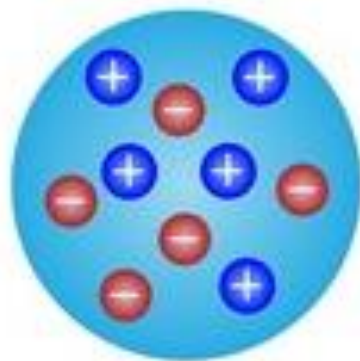
Thomson Model
1897

atoms are NOT
indivisible particles

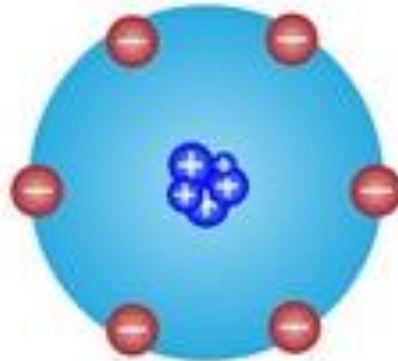
Bohr Model
1922



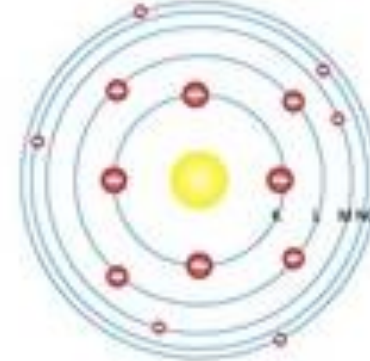
JOHN DALTON, 1803



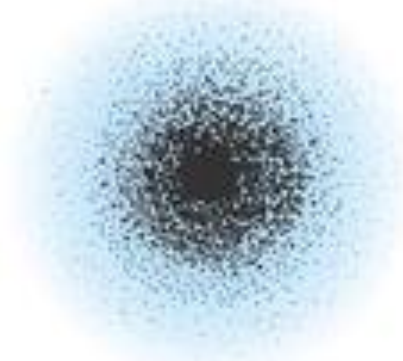
J.J. THOMSON, 1904



ERNEST RUTHERFORD, 1911



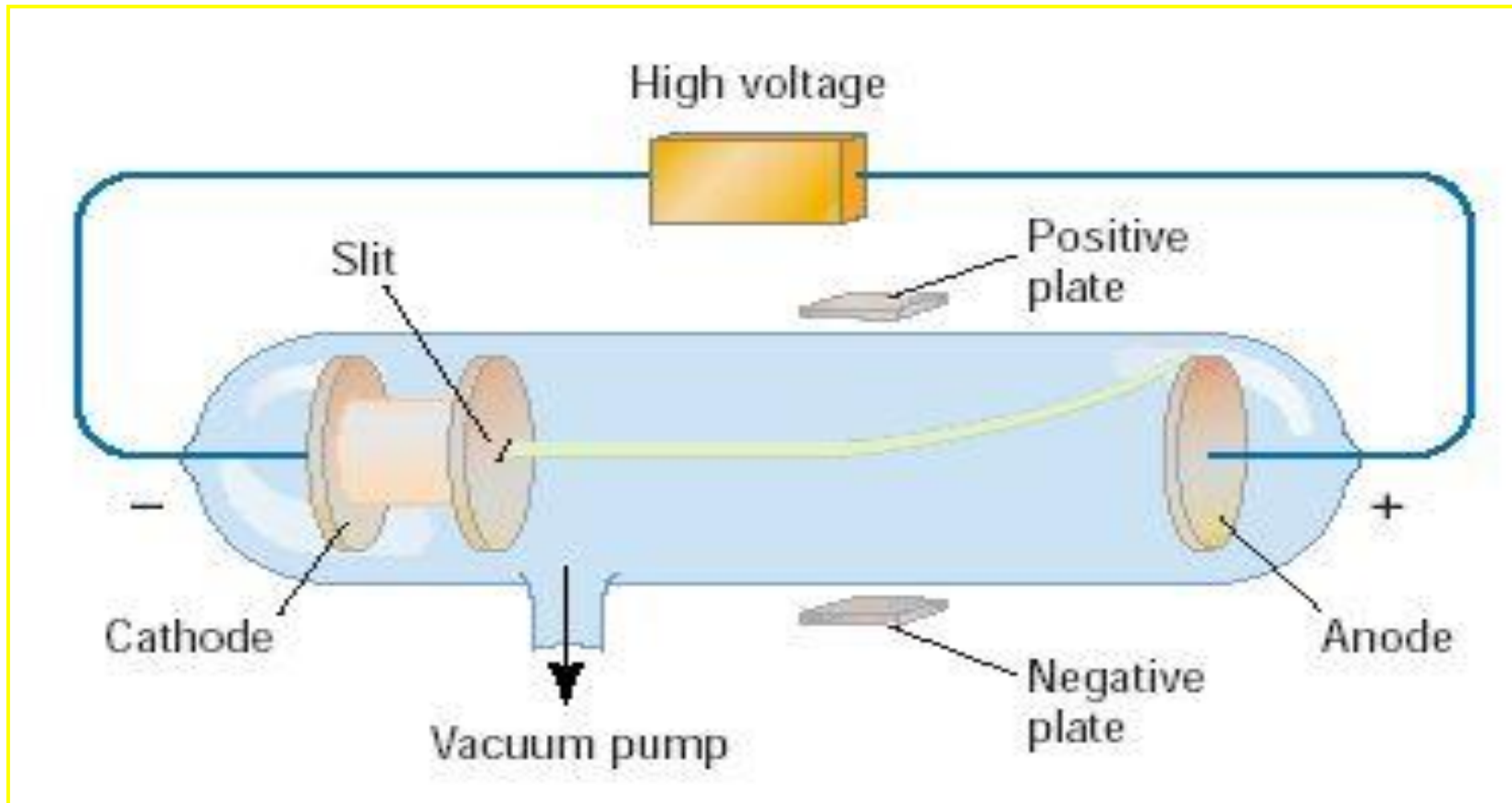
NIELS BOHR, 1913



ERWIN SCHRÖDINGER, 1926

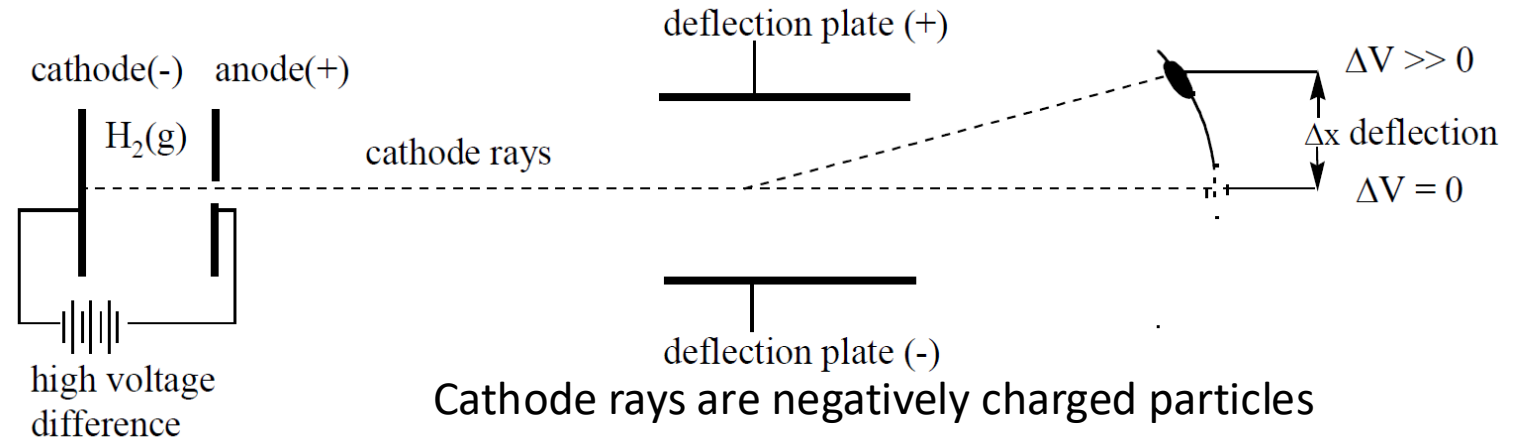
Discovery of the Electron

In 1897, J.J. Thomson used a cathode ray tube to deduce the presence of a negatively charged particle: the **electron**



Discovery of electron: JJ Thomson (1897)

The experiment



If ΔV (the voltage difference between the two deflection plates) = 0 there is no deflection, but when ΔV is $\gg 0$, there is Δx deflection towards the positive plate.

$$\Delta x(-) = \frac{q(-)}{m(-)}$$

$$\Delta x(+) = \frac{q(+)}{m(+)}$$

This negative particle from the cathode ray tubes was later named the **electron** and its mass was later determined to be very small ($m = 9.11 \times 10^{-31} \text{ kg}$).

Atoms are NOT indivisible!

$$\frac{\Delta x(-)}{\Delta x(+)} = \frac{q(-)}{m(-)} / \frac{q(+)}{m(+)} = \frac{m(+)}{m(-)}$$

$$m(-) \lll m(+)$$

Conclusions from the Study of the Electron:

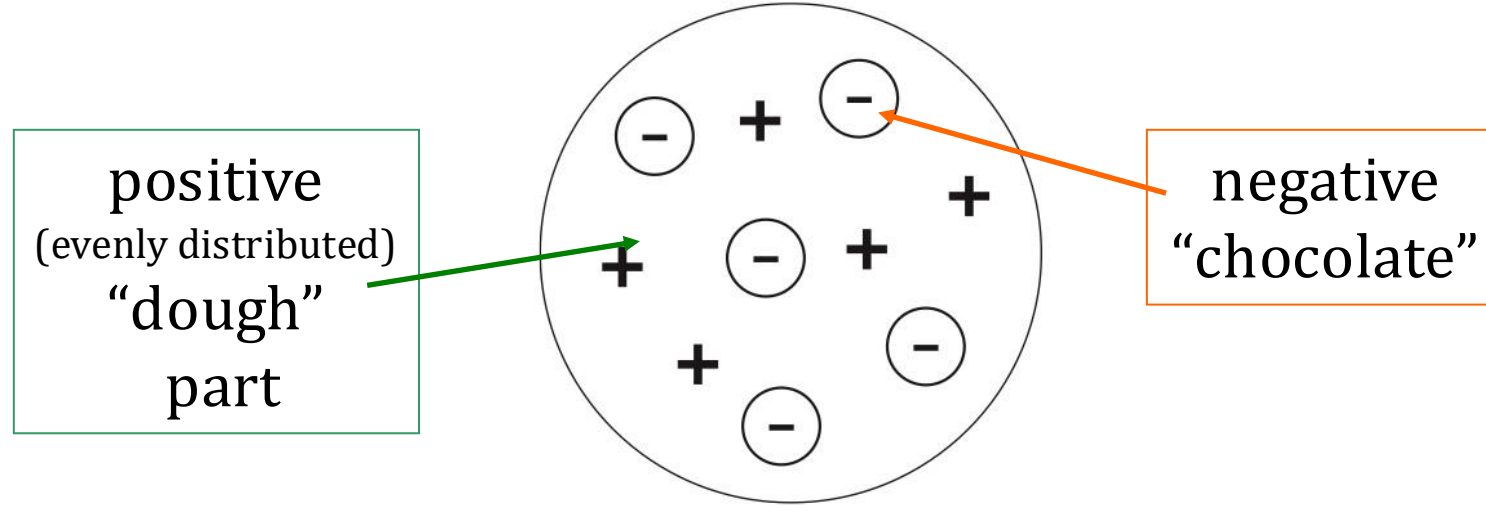
Cathode rays have identical properties regardless of the element used to produce them. All elements must contain identically charged electrons.

Atoms are neutral, so there must be positive particles in the atom to balance the negative charge of the electrons.

Electrons have so little mass that atoms must contain other particles that account for most of the mass.

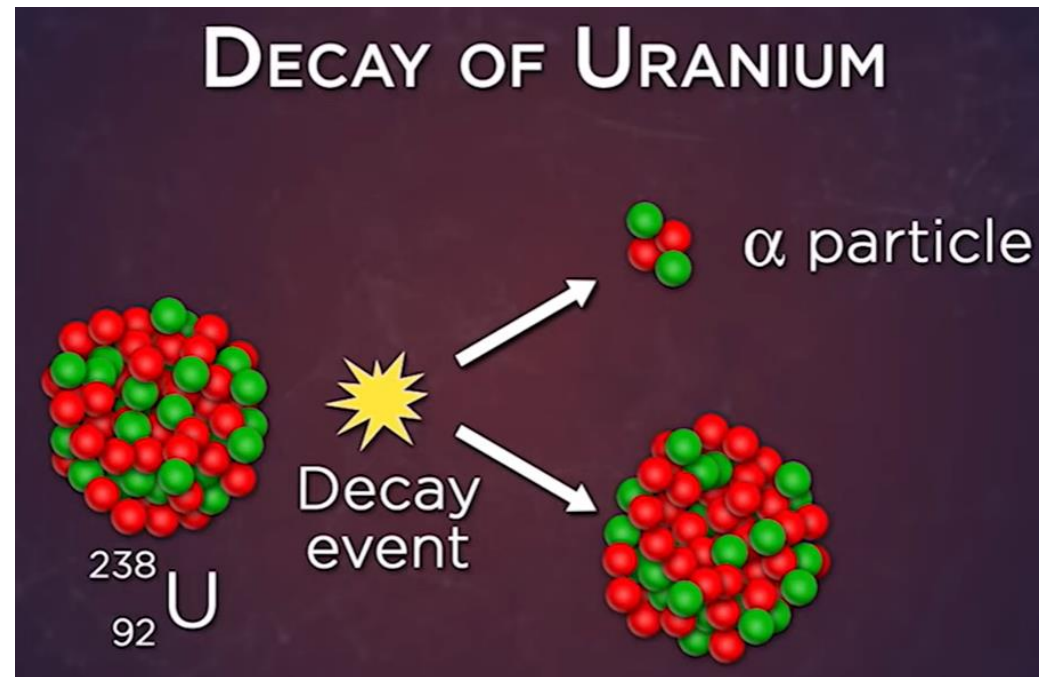
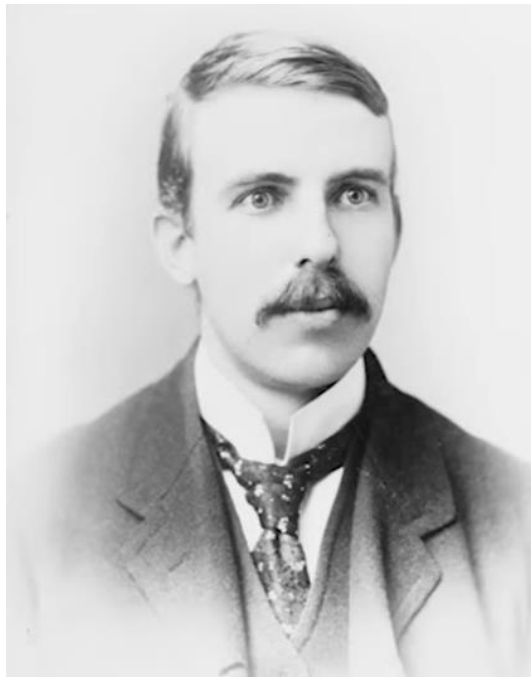
Thomson: “Plum Pudding” or “Chocolate Chip Cookie” Model

- using available data on the atom, **J.J. Thomson** came up with the idea of having charges embedded with Dalton’s Billiard Balls
- Also used cathode ray experiment to discover the existence of the electron



Nuclear Model - 1911

- Ernest Rutherford discovered a huge flaw in the previous concept of the atom during his now famous gold foil experiment



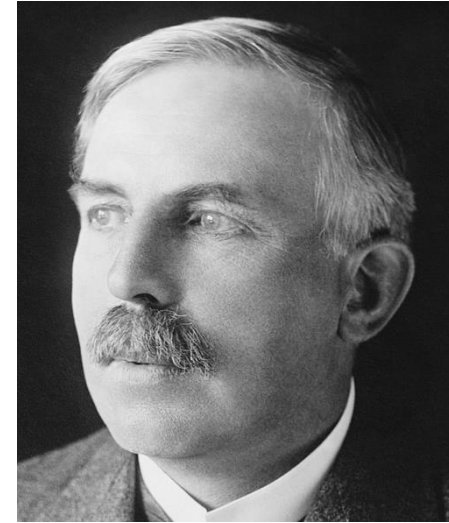
DISCOVERY OF THE NUCLEUS

Ernest Rutherford (Cambridge University, England, 1871-1937) studied α emission from newly-discovered radioactive elements that he got from his friend Marie Curie.



Marie Curie
(France)

Radium Bromide
→
Shipped to Rutherford
for experiment from
France to England



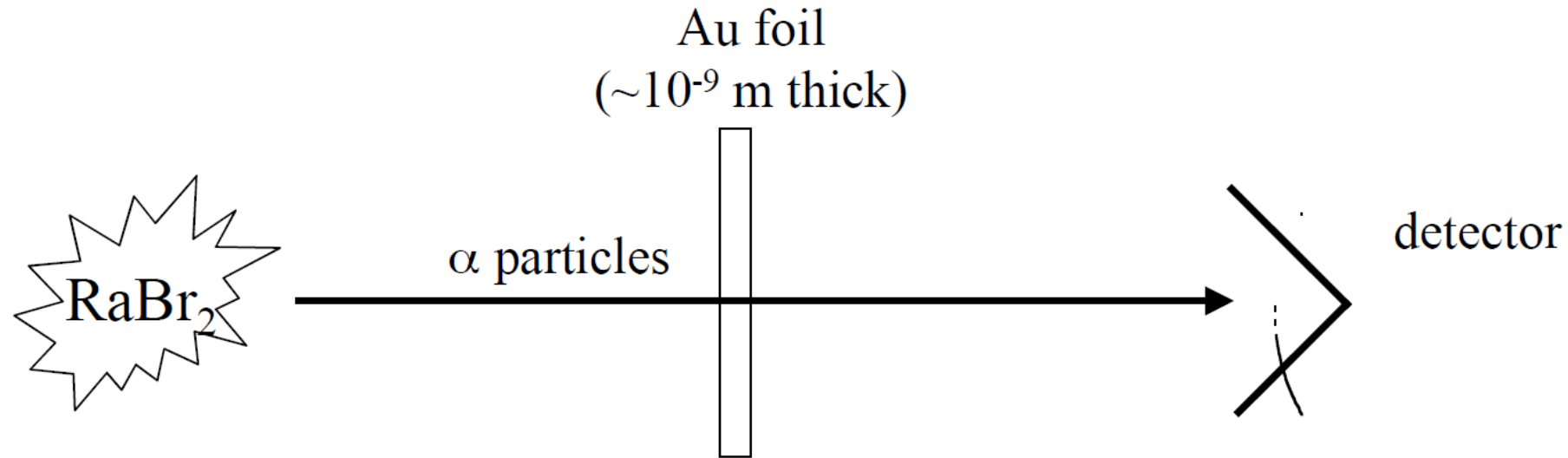
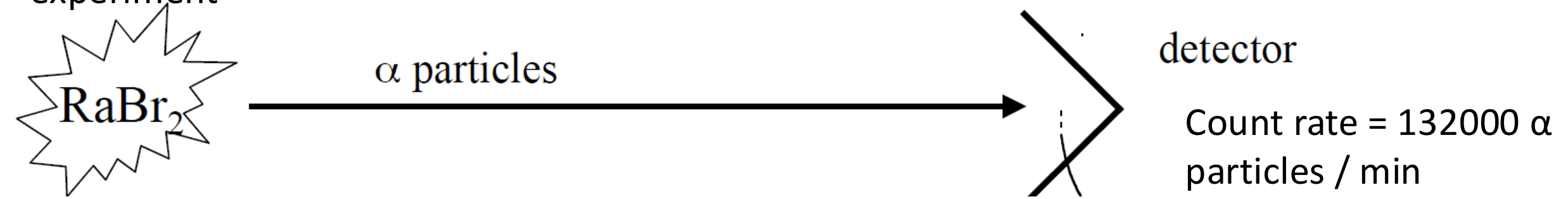
Ernest
Rutherford
(England)

Radium Bromide

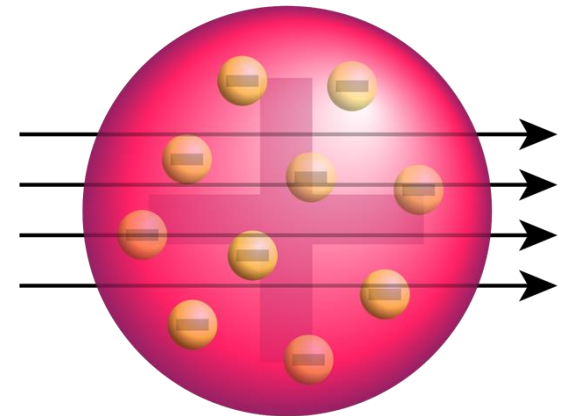


α particles

With postdocs Hans Geiger and undergraduate E. Marsden, the Rutherford Lab did the experiment



Count rate = 132000 α particles / min with or without gold foil
Conclusion – Atom is mostly empty



Little advance experiment from the Rutherford Laboratory: Now the detector is all around for detection of back scatter alpha particles

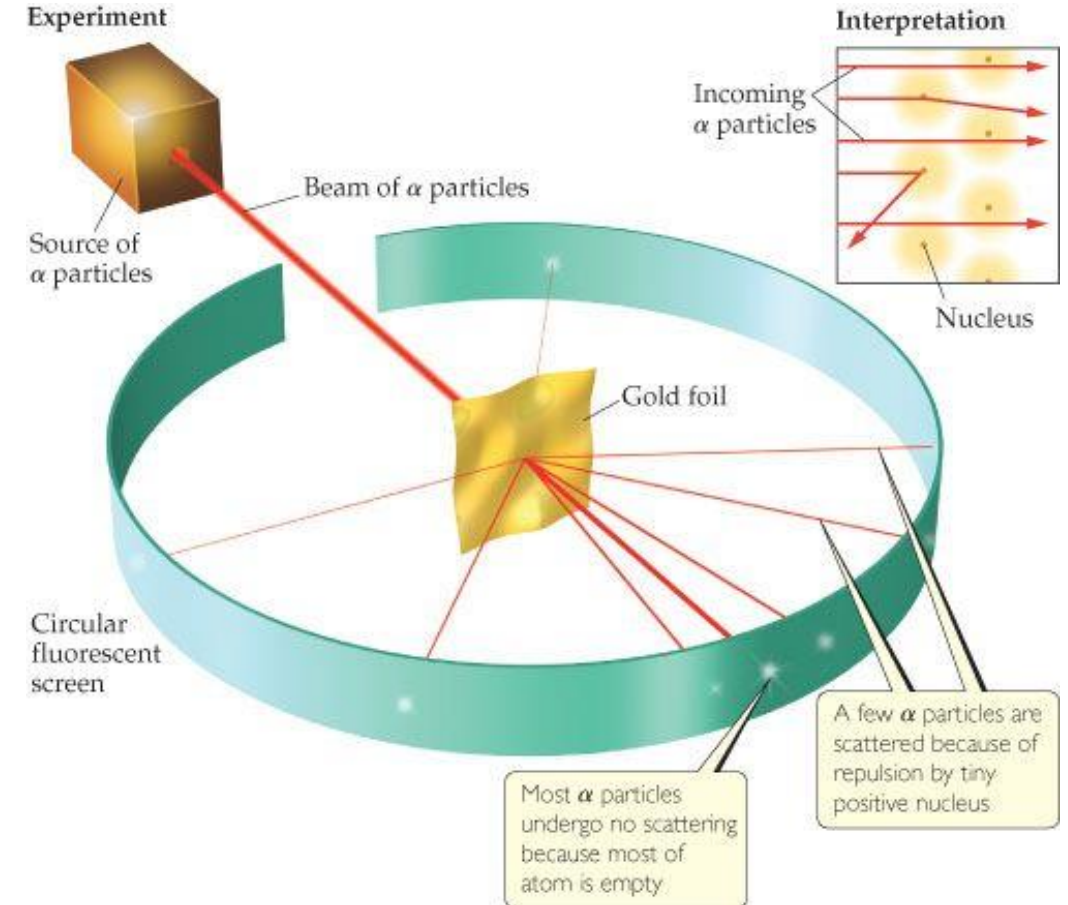
Mass of alpha particle: 8000 electron mass

Speed: 2×10^7 m/s

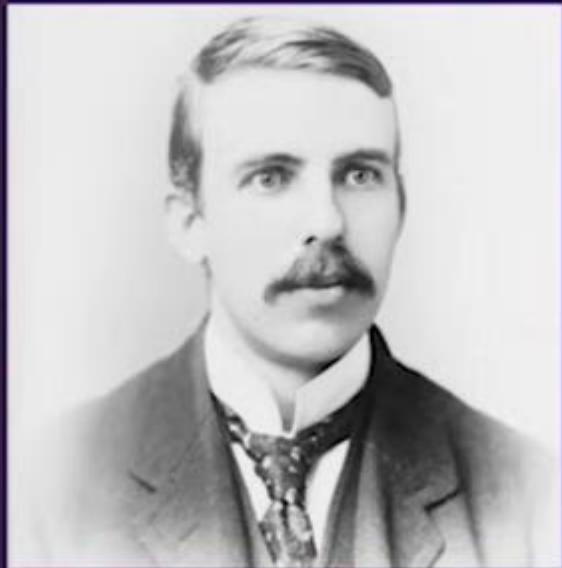
Count rate on backscatter side = 20 α particles / min

Probability (P) of backscattering:

$$P = \frac{\text{count rate backscattered}}{\text{count rate of incident particles}} = \frac{20}{132000} \\ = 2 \times 10^{-4} \text{ (0.02\%)}$$



“It was quite the most incredible event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you!”



Rutherford said – “it was about as incredible as if you had fired a 15 inch shell at a piece of tissue paper and it came back and hit you.”

Interpretation:

- The Gold atoms are mostly empty.
- A very tiny percentage of α particles hit something massive in the atom and backscatter (bounce back). This observation indicates that **most of the mass of the atom is concentrated in a very small volume relative to the volume of the entire atom**. We now call this the NUCLEUS.

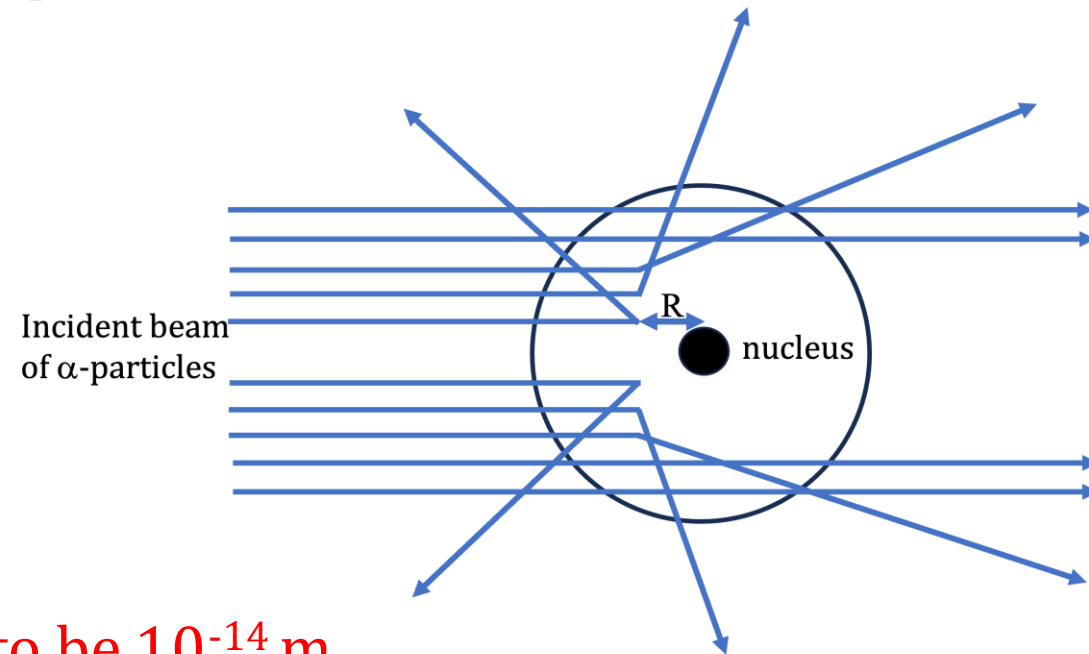
Rutherford proposed the charge on the nucleus to be positive, since electrons are negatively charged and atoms are neutral.

Charge of the electrons in an atom = $-Ze$

Z = atomic number

e = absolute value of electron's charge

Charge of the nucleus of an atom = positive = $+Ze$



- Rutherford calculated the diameter of the nucleus to be 10^{-14} m.

His calculation related the probability of backscattering to the diameter of the nucleus based on the size of the atom and the thickness of the foil.

Rutherford Scattering formula (**Annexure of Chapter 4 – PLEASE DO. It will be taught in detail by Prof. Shandilya**)

The formula that Rutherford obtained for alpha particle scattering by a thin foil on the basis of the nuclear model of the atom is

Rutherford
Scattering Formula

$$N(\theta) = \frac{N_i n t Z^2 e^4}{(8\pi\epsilon_0)^2 r^2 K E^2 \sin^4\left(\frac{\theta}{2}\right)}$$

$N(\theta)$ = number of alpha particles per unit area that reach the screen at a scattering angle of θ

N_i = total number of alpha particle that reach the screen

n = number of atoms per unit volume in the foil

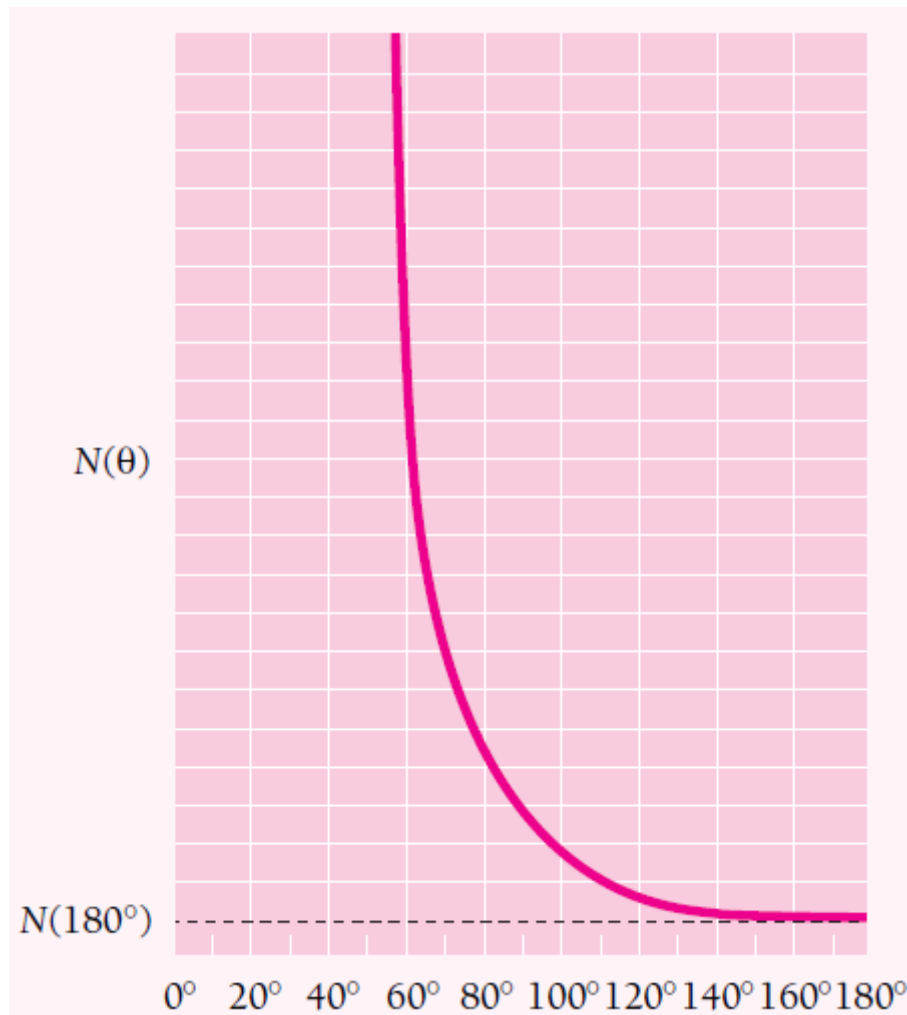
Z = atomic number of the foil atoms

r = distance of the screen from the foil

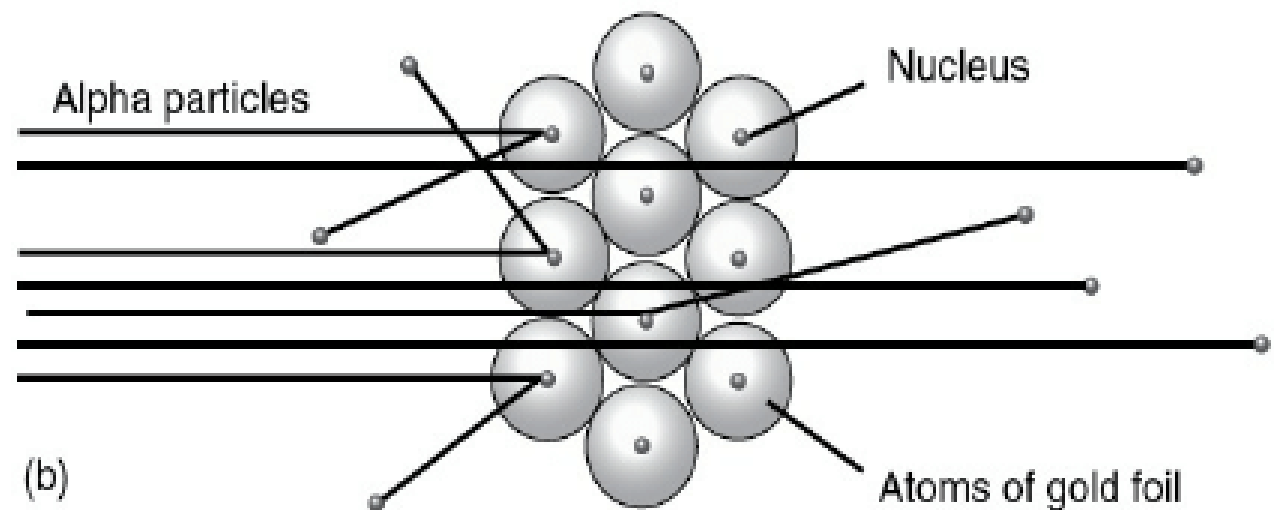
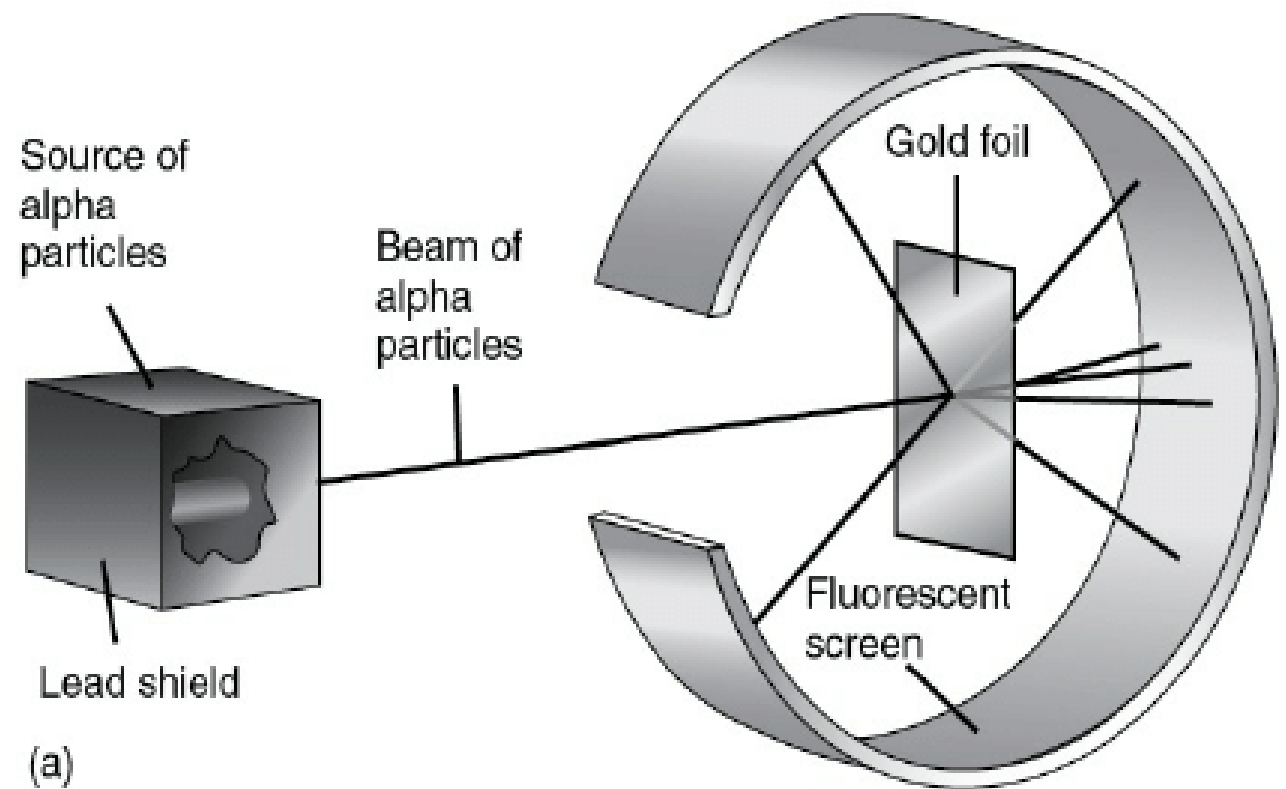
KE = kinetic energy of the alpha particles

t = foil thickness

The predictions of Scattering Formula agreed with the measurements of Geiger and Marsden, which supported the hypothesis of the nuclear atom.



Because $N(\theta)$ is inversely proportional to $\sin^4(\theta/2)$ the variation of $N(\theta)$ with θ is very pronounced : only 0.14 percent of the incident alpha particles are scattered.



Nuclear Dimensions

Rutherford assumed $\rightarrow$ the size of a target nucleus $<$ the minimum distance R (the distance the incident alpha particles approach the nucleus before being deflected away)

R = The distance of **closest approach** R for the most energetic alpha particles.

Rutherford scattering $\rightarrow$ gives us a way to find an upper limit to nuclear dimensions.

An alpha particle will have its **smallest** R when **it approaches a nucleus head on**, which will be followed by a **180° scattering**.

At the instant of closest approach the initial kinetic energy (KE) of the particle is entirely converted to electric potential energy, and so at that instant

$$KE_{initial} = PE = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{R}$$

since the charge of the alpha particle is $2e$ and that of the nucleus is Ze .

Distance of closest approach

$$R = \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{KE_{initial}}$$

The maximum KE found in alpha particles of natural origin is 7.7 MeV,

$$7.7 \text{ MeV} = 1.2 \times 10^{-12} \text{ J}$$

$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$\begin{aligned} R &= \frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{KE_{\text{initial}}} \\ &= \frac{9 \times 10^9 \text{ Nm}^2}{\text{C}^2} \frac{(2Z)(1.6 \times 10^{-19} \text{ C})^2}{1.2 \times 10^{-12} \text{ J}} \\ &= 3.8 \times 10^{-16} Zm \end{aligned}$$

The atomic number of gold is $Z=79$

$$R = 3.0 \times 10^{-14} m$$

Radius of the gold nucleus $< 3.0 \times 10^{-14} m \rightarrow$ less than 10^{-4} times of the atomic radius

ELECTRON ORBITS

Hydrogen atom : Single electron.

Assume a circular electron orbit for convenience.

The centripetal force

$$F_c = \frac{mv^2}{r}$$

holding the electron in an orbit r from the nucleus is provided by the electric force between them

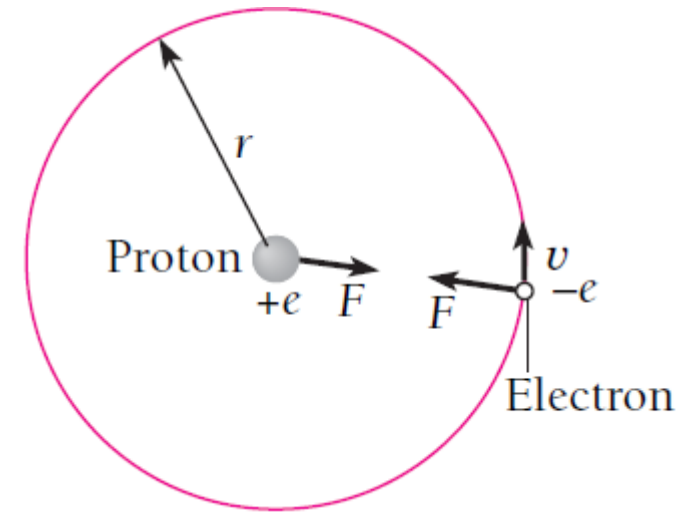
$$F_e = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

The condition for a dynamically stable orbit is

$$F_c = F_e$$

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

Electron velocity, $v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$



The total energy E of the electron in a hydrogen atom is the sum of its kinetic and potential energies, which are

$$\text{KE} = \frac{1}{2}mv^2 \quad \text{PE} = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

The minus sign $\rightarrow$ $\text{PE} = 0$ at $r = \infty$, that is, when the electron and proton are infinitely far apart

$$E = \text{KE} + \text{PE} = \frac{1}{2}mv^2 - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

$$E = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r} - \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

Total energy of hydrogen atom

$$E = -\frac{1}{8\pi\epsilon_0} \frac{e^2}{r}$$

The total energy of the electron is negative.

This holds for every atomic electron and reflects the fact that **it is bound to the nucleus**.

If $E > \text{zero} \rightarrow$ an electron would **not follow a closed orbit around the nucleus**.

(FAILURE OF) THE CLASSICAL DESCRIPTION OF AN ATOM

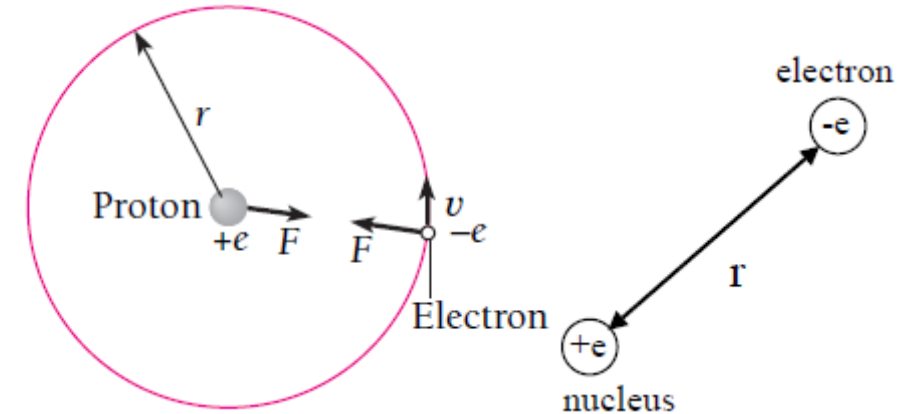
How is an electron bound to the nucleus? What is the force between them? **Coulomb's Force Law** describes the force of attraction and repulsion between charged particles.

$$\text{Coulomb Force } (F_r) = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$$

Q_x = charge on particle x

r = distance between two charges

ϵ_0 = permittivity constant of a vac
($8.854 \times 10^{-12} \text{ C}^2\text{J}^{-1}\text{m}^{-1}$)



When charges are the same sign, acceleration of the particles will push them apart. The force is positive and repulsive.

When charges are the opposite sign, acceleration will pull them together. The force is negative and attractive.

Consider an H atom ($Z=1$) with 1 electron and 1 proton.
When $r \rightarrow \infty$, $F(r) = 0$

If the electron and the nucleus are infinitely attracted, why do they stay apart.

As $r \rightarrow 0$, $F(r) = \infty \rightarrow$ infinitely attractive

Coulomb's law only describes force as a function of r (distance). It doesn't tell us how r changes with time, t .

To write the force as a function of time, we need a classical equation of motion that can describe how the electron and nucleus move under force.

Newton's Second law:

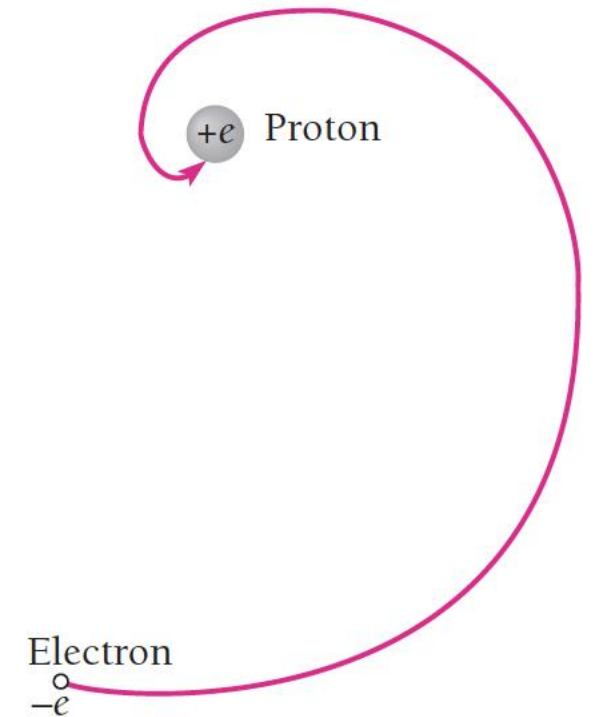
Force, $F = ma$ (mass X acceleration)

$F = m (dv/dt)$ (function of velocity) = $m(d^2r/dt^2)$ (function of distance)

Plug in Coulomb's force law

For $r_{\text{initial}} = 0.5 \times 10^{-10} \text{m}$, calculate how fast it should go to zero. The answer is very fast.

$r=0$ at $t = 10^{-10} \text{s}$



This predicts that the electron should crash into the nucleus in 10^{-10} sec. But why is it not doing that?

So what is the problem here?

1. Coulomb force law?
2. Newton's second law?

So problem here is → The laws of classical Mechanics no longer work at this size scale.

A new kind of mechanics was needed to describe this and other “unsettling” observations.

QUANTUM MECHANICS provides a single and comprehensive theory that explains the behavior of matter on an atomic (nanometer or smaller) scale.

Atomic Spectra

Each element has a characteristic line spectrum

ABSORPTION AND EMISSION SPECTRA

400 nm 450 nm 500 nm 550 nm 600 nm 650 nm 700 nm

SODIUM

Absorption

Emission

NITROGEN

Absorption

Emission

HYDROGEN

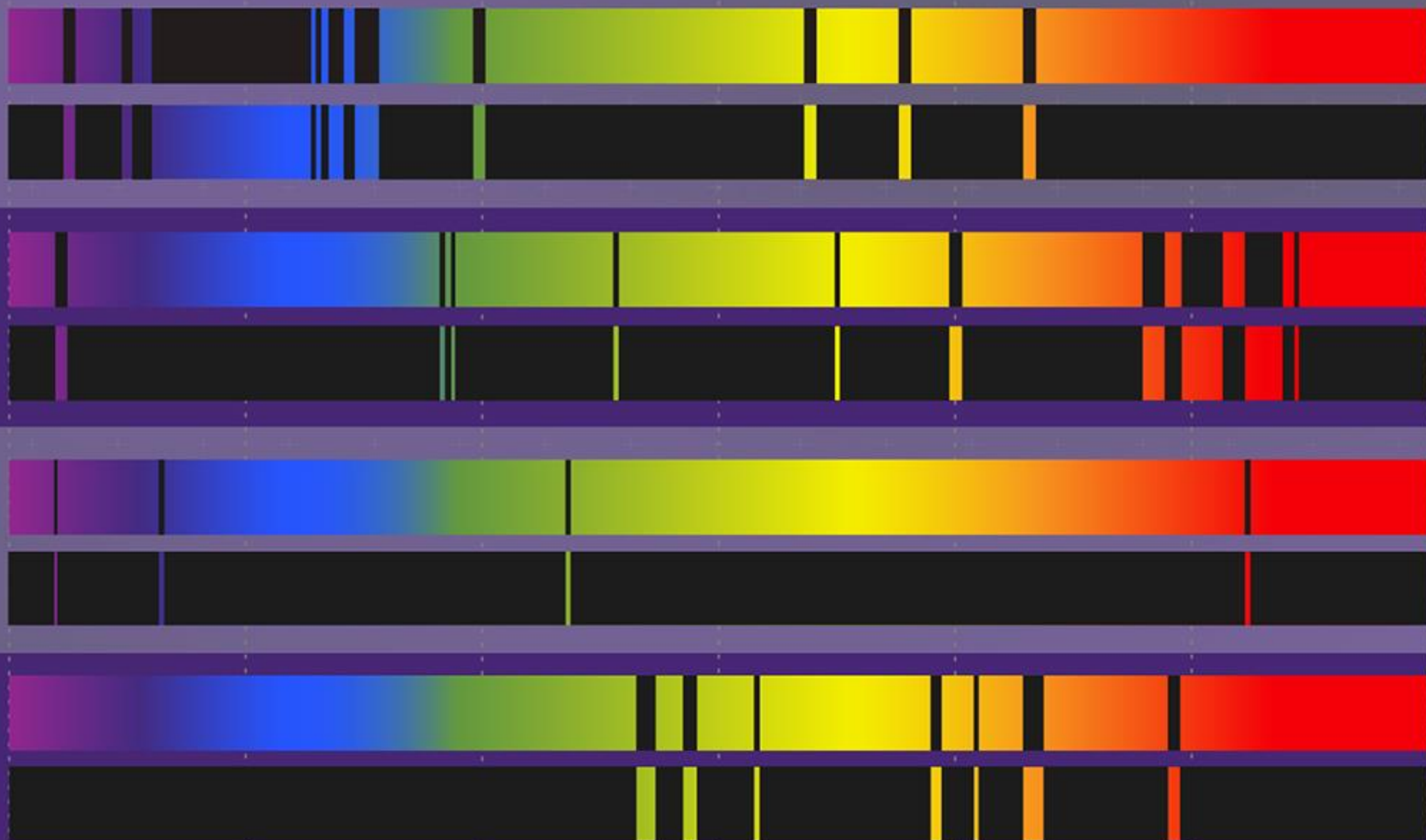
Absorption

Emission

OXYGEN

Absorption

Emission



Early quantum ideas and the Bohr Model

Bohr's postulates were based on the observation of discrete spectra and the results of the α -scattering experiment of Rutherford.

The postulates were

- **Electrons move around the nucleus in discrete orbits.** These orbits correspond to fixed energies and are stable (i.e. no radiation is emitted by such electrons)
- **Radiation is emitted only when the electron jumps from one discrete orbit to another orbit of a lower energy.** When electrons absorb radiation, the reverse transition occurs.
- **A dynamic equilibrium exists between the centrifugal force and the coulomb force between the electron and the nucleus:**

$$\frac{(Ze)^2}{4\pi\epsilon_0 r^2} = m\omega^2 r \rightarrow \omega = \frac{e^2}{4\pi\epsilon_0 m r^3}$$

The electron revolves at the frequency ω in the orbit.

Classical electrodynamics predicts that the charge will radiate at this very frequency and lose energy. Bohr's postulates override this.

The total energy of the electron is

$$\frac{(Ze)^2}{4\pi\epsilon_0 r^2} = m\omega^2 r \rightarrow \omega = \frac{e^2}{4\pi\epsilon_0 m r^3}$$

$$E = E_{kin} + E_{pot} = \frac{1}{2} m \omega^2 r^2 - \frac{e^2}{4\pi\epsilon_0 r}$$

Using the above obtained value of ω , we get

$$E = -\frac{e^2}{8\pi\epsilon_0 r} = \frac{1}{2} \frac{\omega^{2/3}}{(4\pi\epsilon_0)^{2/3}} (me^4)^{1/3}$$

Bohr's important insight was to recognize that the frequency of radiation (according to the postulates) should be given by

$$h\nu = E_n - E_{n'}$$

where $h\nu$ is the radiation quantum as proposed by Planck and that this radiation must be the same as that.

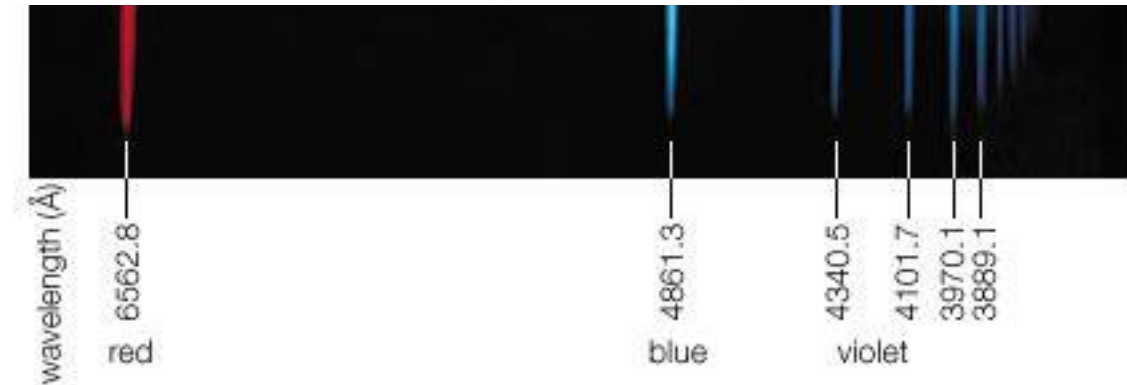
Balmer $\frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \quad n = 3, 4, 5, 6 \dots$ $R \rightarrow$ **Rydberg constant**

Rydberg constant $R = 1.097 \times 10^7 \text{ m}^{-1} = 0.01097 \text{ nm}^{-1}$

H_α line $\rightarrow n = 3, \rightarrow 656.3\text{nm}$ corresponds to 1.9eV

H_β line $\rightarrow n = 4$, and so on.

The series limit corresponds to $n = \infty$, at $\lambda = \frac{4}{R}$, in agreement with experiment and corresponds to 364.7nm $\rightarrow$ 3.4eV.



Balmer series $\rightarrow$ wavelengths in the visible portion of the hydrogen spectrum.

Lyman series $\rightarrow$ wavelengths in the ultraviolet region $\rightarrow$ smallest is 91.2 nm $\rightarrow$ 13.598eV

Lyman $\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{n^2} \right) \quad n = 2, 3, 4, \dots$

Three spectral series have been found in the infrared region.
The wavelengths specified by the formulas

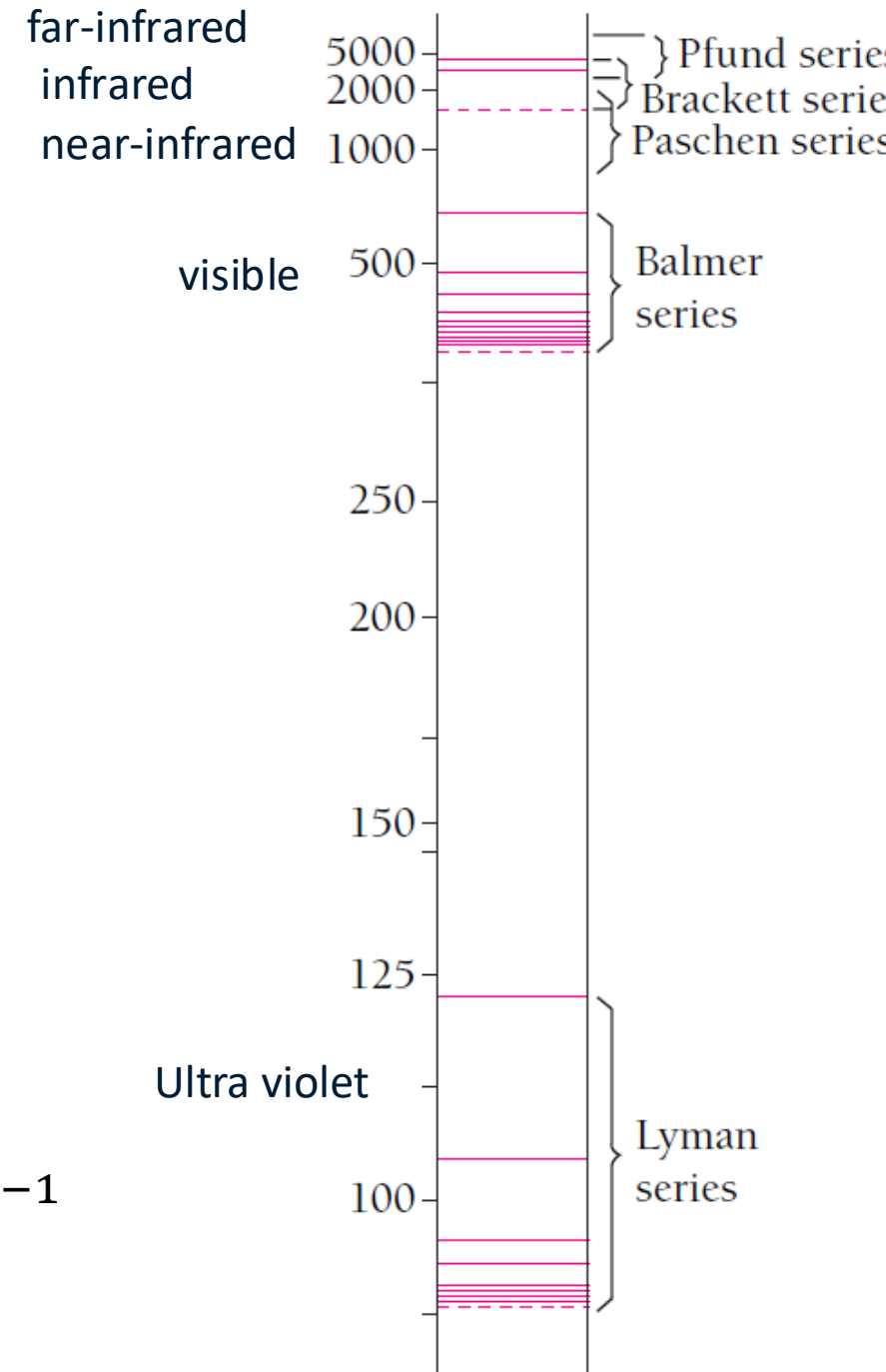
Paschen $\frac{1}{\lambda} = R \left(\frac{1}{3^2} - \frac{1}{n^2} \right) \quad n = 4, 5, 6 \dots$

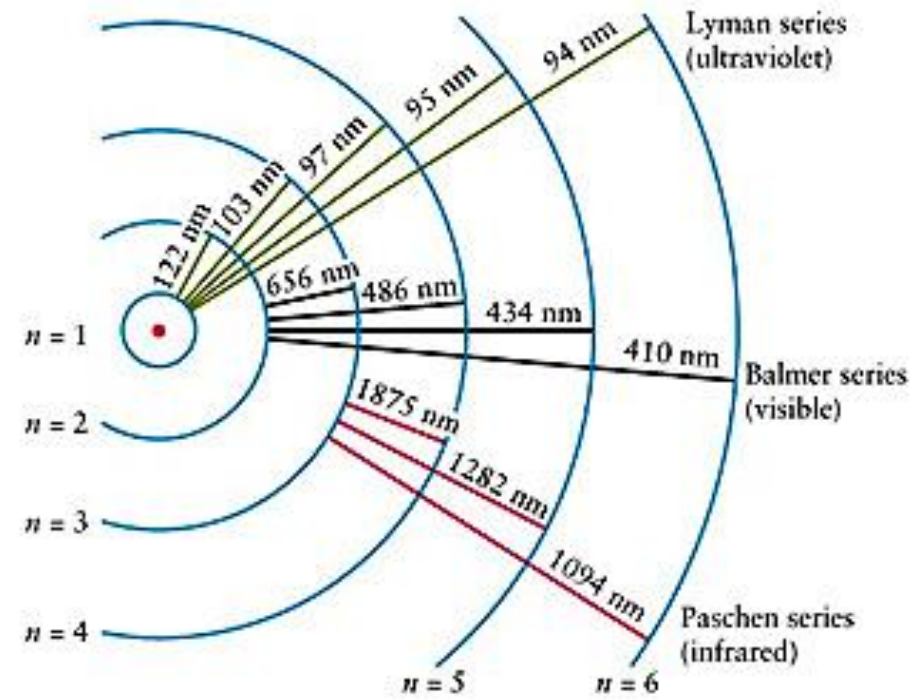
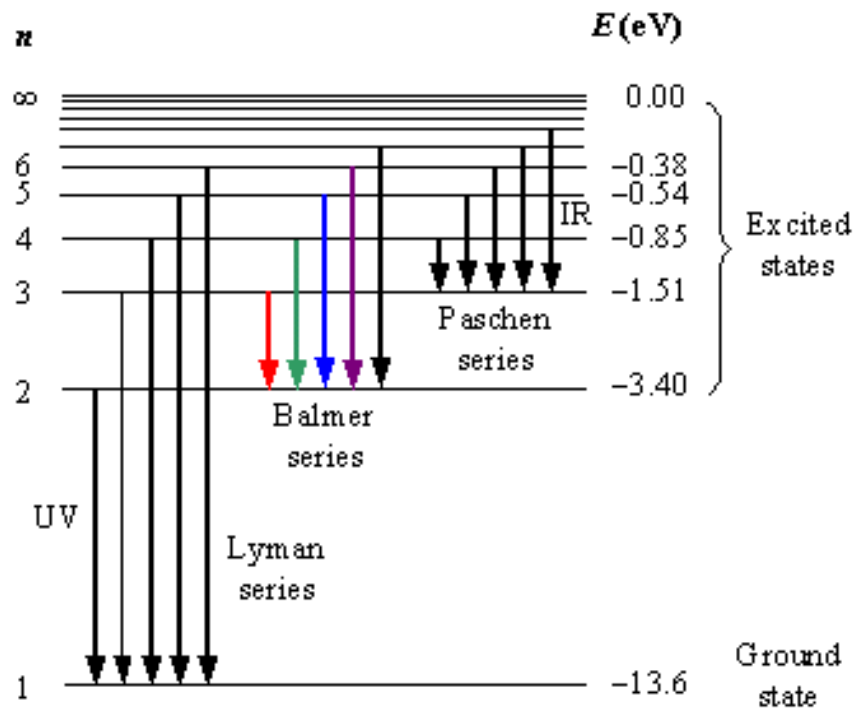
Brackett $\frac{1}{\lambda} = R \left(\frac{1}{4^2} - \frac{1}{n^2} \right) \quad n = 5, 6, 7 \dots$

Pfund. $\frac{1}{\lambda} = R \left(\frac{1}{5^2} - \frac{1}{n^2} \right) \quad n = 6, 7, 8 \dots$

The value of R is the same in all the spectral series.

Rydberg constant $R = 1.097 \times 10^7 \text{ m}^{-1} = 0.01097 \text{ nm}^{-1}$





Hydrogen Series of Spectral Lines $\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{k^2} \right)$

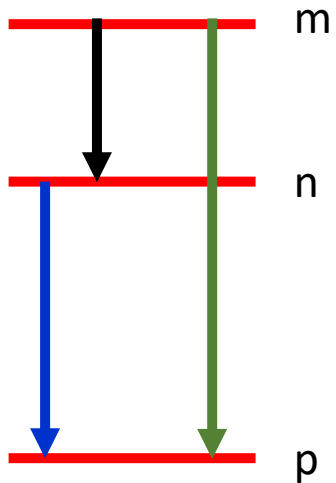
Discoverer (year)	Wavelength	n	k
Lyman (1916)	Ultraviolet	1	>1
Balmer (1885)	Visible, ultraviolet	2	>2
Paschen (1908)	Near Infrared	3	>3
Brackett (1922)	Infrared	4	>4
Pfund (1924)	Far Infrared	5	>5

Rydberg-Ritz Combination Principle

The wavenumber of any spectral line in an atom's spectrum can be expressed as the **sum or difference of the wavenumbers** of **two other spectral lines** from the **same element**

Transition from m^{th} to n^{th} level

$$h\nu_{mn} = E_m - E_n$$



Transition from n^{th} to p^{th} level

$$h\nu_{np} = E_n - E_p$$

Transition from m^{th} to p^{th} level

$$h\nu_{mn} + h\nu_{np} = E_m - E_n + E_n - E_p = E_m - E_p = h\nu_{mp}$$

Que: Experiments indicate that 13.6 eV is required to separate a hydrogen atom into a proton and an electron; that is, its total energy is $E = -13.6 \text{ eV}$. Find the orbital radius and velocity of the electron in a hydrogen atom.

$$13.6 \text{ eV} = 2.2 \times 10^{-18} \text{ J}$$

$$E = -\frac{e^2}{8\pi\epsilon_0 r}$$

$$\begin{aligned} r &= -\frac{e^2}{8\pi\epsilon_0 E} = -\frac{(1.6 \times 10^{-19} \text{ C})^2}{(8\pi)(8.85 \times 10^{-12} \text{ F/m})(-2.2 \times 10^{-18} \text{ J})} \\ &= 5.3 \times 10^{-11} \text{ m} \end{aligned}$$

BOHR'S RADIUS

$$\begin{aligned} v &= \frac{e}{\sqrt{4\pi\epsilon_0 m r}} = \frac{1.6 \times 10^{-19} \text{ C}}{\sqrt{(4\pi)(8.85 \times 10^{-12} \text{ F/m})(9.1 \times 10^{-31} \text{ kg})(5.3 \times 10^{-11} \text{ m})}} \\ &= 2.2 \times 10^6 \text{ m/s} \end{aligned}$$

THE BOHR ATOM

The de Broglie wavelength of an electron is $\lambda = \frac{h}{mv}$

v = electron velocity

$$v = \frac{e}{\sqrt{4\pi\epsilon_0 mr}}$$

Orbital electron
wavelength

$$\lambda = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$$

Radius of electron orbit, $r = 5.3 \times 10^{-11} \text{ m}$

The electron wavelength

$$\lambda = \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{1.6 \times 10^{-19} \text{ C}} \sqrt{\frac{(4\pi)(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(5.3 \times 10^{-11} \text{ m})}{9.1 \times 10^{-31} \text{ kg}}}$$
$$= 33 \times 10^{-11} \text{ m}$$

This wavelength is exactly the same as the circumference of the electron orbit,

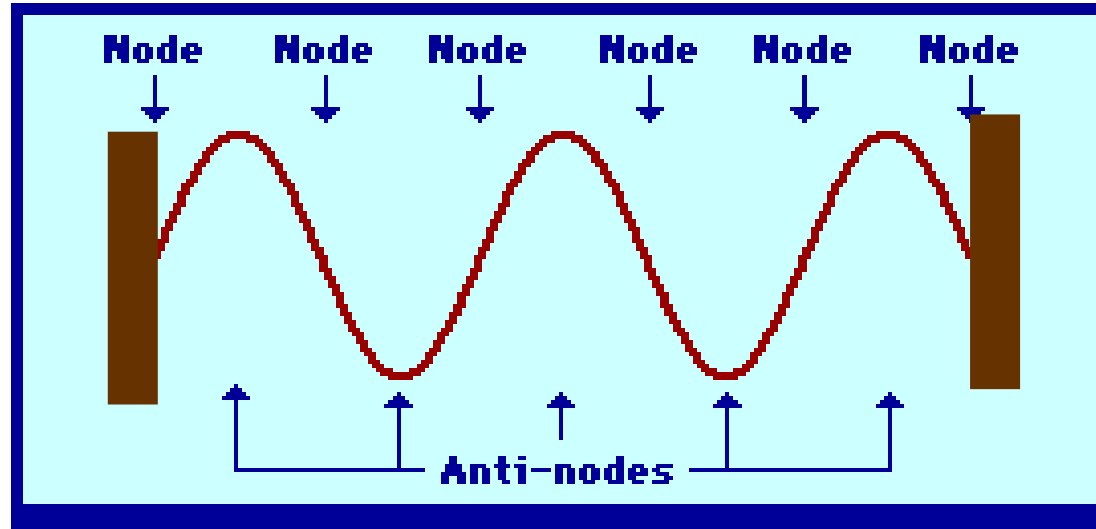
$$2\pi r = 33 \times 10^{-11} \text{ m}$$

What makes Bohr's allowed energy levels allowed?

Recall what happens when we force
waves into confined spaces:



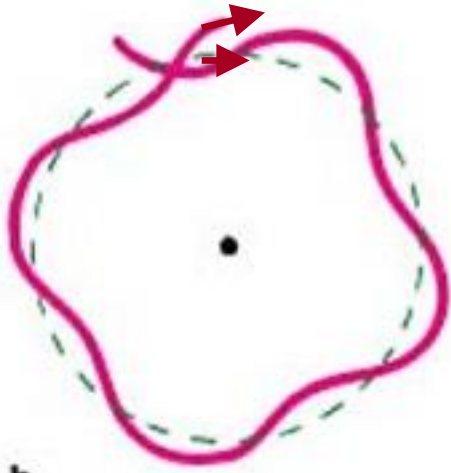
Confined waves



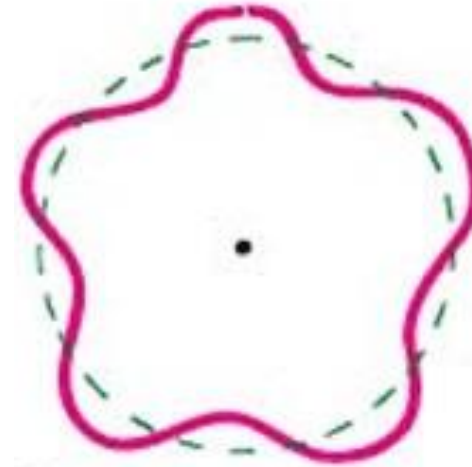
Only waves with wavelengths that **just fit in** survive
(all others cancel themselves out)

Electrons in atoms are confined matter waves

This wave, as it goes around, will interfere with itself destructively and cancel itself out



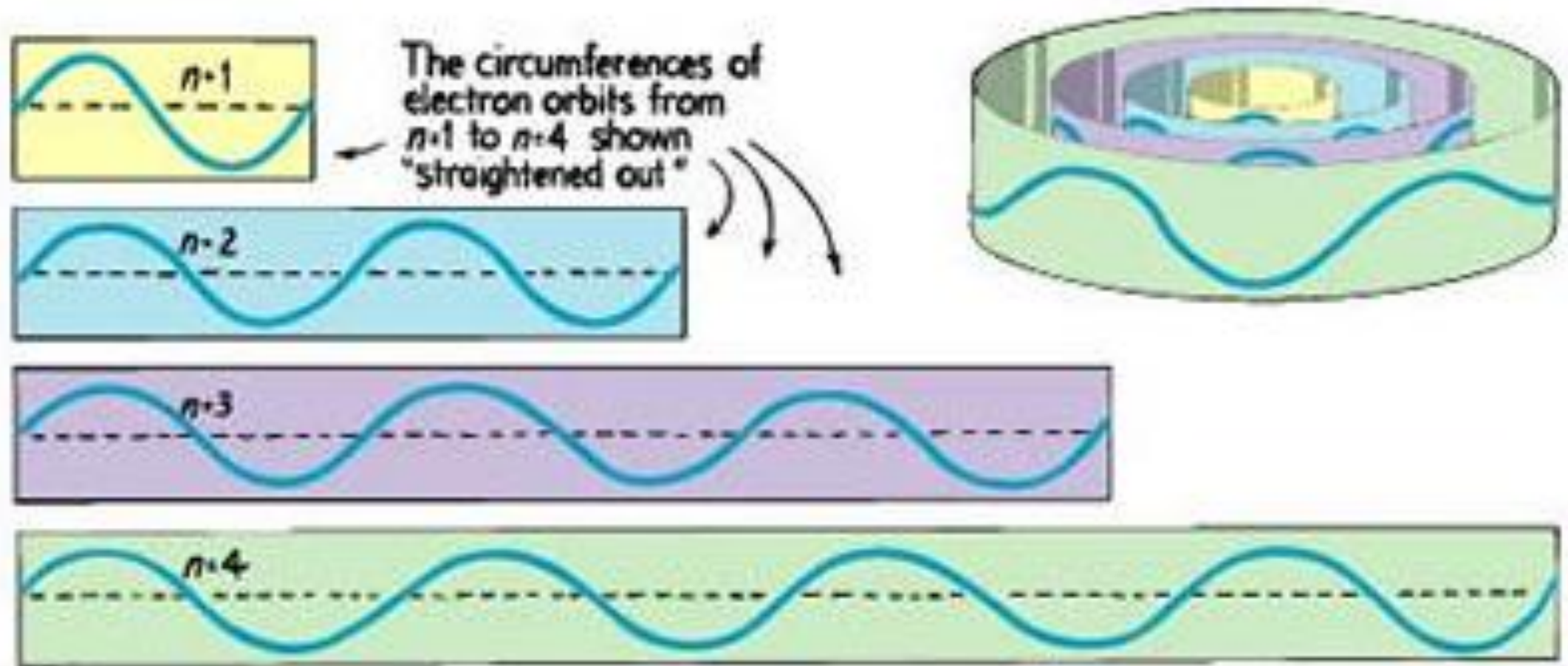
However, if the circumference is exactly an integer number of wavelengths, successive turns will interfere constructively



Bohr's allowed energy states correspond to those with orbits that are integer numbers of wavelengths

Bohr orbits

An electron can circle a nucleus only if its orbit contains an integral number of *de Broglie* wavelengths



Condition for orbit stability

$$n\lambda = 2\pi r_n \quad n = 1, 2, 3 \dots$$

r_n = the radius of the orbit that contain n wavelengths.

n = the **quantum number** of the orbit.

Substituting for λ , the electron wavelength

$$\frac{nh}{e} \sqrt{\frac{4\pi\epsilon_0 r_n}{m}} = 2\pi r_n$$

Orbital electron wavelength $\lambda = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$

The possible electron orbits are those whose radii



$$r_n = \frac{n^2 h^2 \epsilon_0}{\pi m e^2} \quad n = 1, 2, 3, \dots$$

The innermost radius is called Bohr Radius

Bohr radius

$$a_0 = r_1 = 5.292 \times 10^{-11} \text{ m}$$

The other radii in terms of Bohr Radius are

$$r_n = n^2 a_0$$

ENERGY LEVELS AND SPECTRA

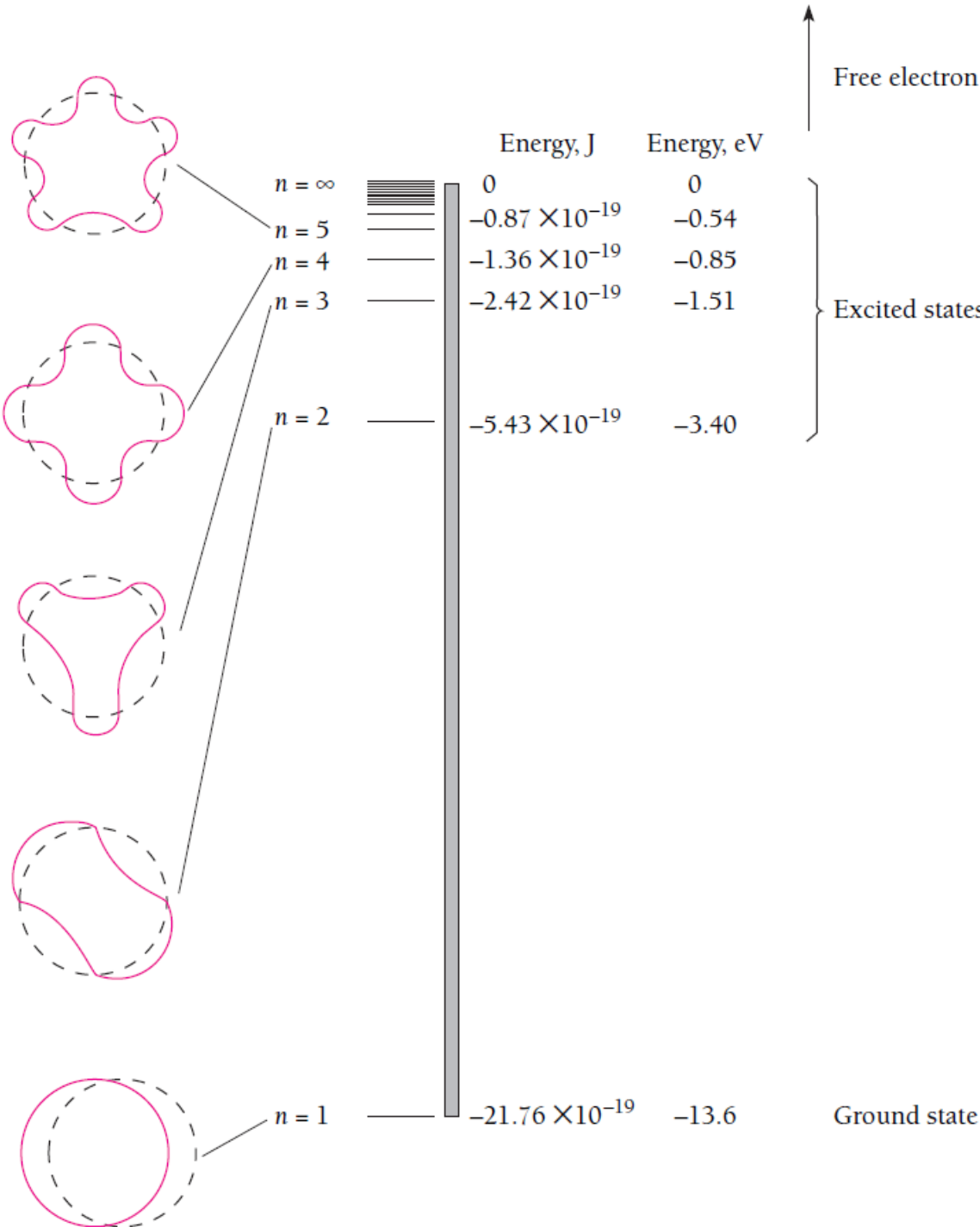
The electron energy E_n is given in terms of the orbit radius r_n as

$$E_n = -\frac{e^2}{8\pi\epsilon_0 r_n}$$

The **energy levels** of the hydrogen atom

$$E_n = -\frac{me^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2}\right) = \frac{E_1}{n^2} \quad n = 1, 2, 3, \dots$$

$$E_1 = -2.18 \times 10^{-18} \text{ J} = -13.6 \text{ eV}$$



Bohr's Correspondence principal The greater the $Q^\#$, the closer Quantum
proaches classical mechanics.

In limit that $n \rightarrow \infty$,
quantum mechanics must agree with classical physics

$$E_{\text{photon}} = 13.6 \text{ eV} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) = hf_{\text{photon}}$$

$$n_i \approx n_f \text{ when both } n_i, n_f \gg 1$$

Bohr's correspondence principle considers transitions between **nearby very-high- n states**,
for example

$$n_i = n, n_f = n - 1,$$

with

$$n \rightarrow \infty.$$

Although n and $n - 1$ always differ by exactly 1, their **relative difference**

$$\frac{n_i - n_f}{n_i} = \frac{1}{n}$$

goes to zero as $n \rightarrow \infty$.

The Bohr energy levels are

$$E_n = -\frac{13.6}{n^2} \text{ eV}.$$

At small n , the levels are widely separated. For example,

$$E_1 = -13.6 \text{ eV}, E_2 = -3.4 \text{ eV}.$$

But at very large n ,

$$E_n - E_{n-1}$$

becomes extremely small. The energy levels start looking almost continuous.

For a transition

$$n \rightarrow n - 1,$$

the emitted photon has

$$hf = 13.6 \left[\frac{1}{(n-1)^2} - \frac{1}{n^2} \right].$$

For very large n , we can approximate

$$\frac{1}{(n-1)^2} \simeq \frac{1}{n^2} \left(1 + \frac{2}{n} \right),$$

so

$$\frac{1}{(n-1)^2} - \frac{1}{n^2} \simeq \frac{2}{n^3}.$$

Therefore,

$$hf \simeq \frac{27.2 \text{ eV}}{n^3}.$$

The beautiful part of the correspondence principle → If you calculate the **classical orbital frequency** of the electron in the n -th Bohr orbit,

$$f_{\text{orbit}} = \frac{v_n}{2\pi r_n},$$

v_n = speed of the electron in the n -th Bohr orbit.

you obtain exactly the same large- n result as the photon frequency:

$$f_{\text{photon}} \rightarrow f_{\text{orbit}}$$

for

$$\boxed{n \rightarrow \infty, \Delta n = 1}.$$

For very large quantum numbers, transitions between neighbouring quantum states produce radiation whose frequency approaches the classical orbital frequency of the electron.

This is Bohr's correspondence principle:

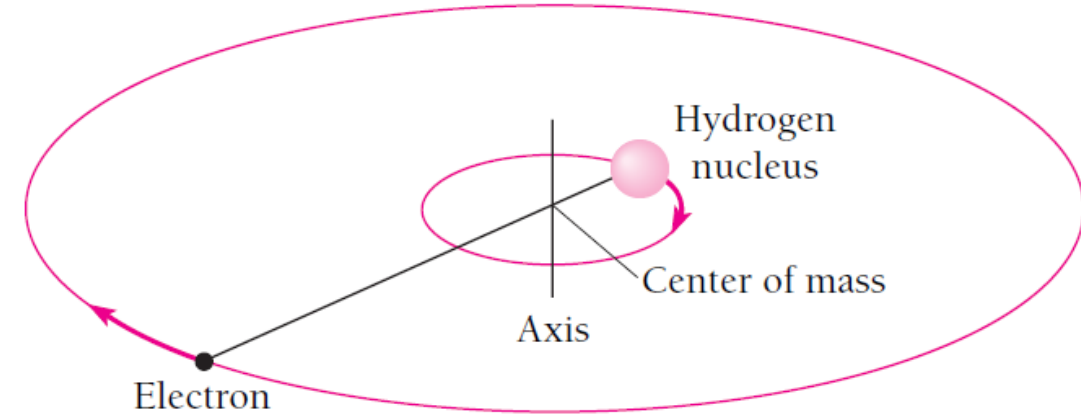
Quantum physics $\rightarrow$ classical physics in the appropriate large-quantum-number limit.

$n = 1, 2, 3$	:	levels clearly discrete
$n = 100, 101, 102$	:	levels extremely close
$\downarrow$		
$n \rightarrow \infty$	:	almost a continuum $\rightarrow$ classical behaviour

NUCLEAR MOTION

The nuclear mass affects the wavelengths of spectral lines

- So far, we've assumed the hydrogen nucleus (a proton) remains stationary while the electron orbits it.
- In reality, both the nucleus and electron orbit their common center of mass, which lies near the nucleus due to its much greater mass.
- This system is equivalent to a single particle of mass m' orbiting the heavier nucleus.



Reduced mass

$$m' = \frac{mM}{m + M}$$

m = electron mass

M = nuclear mass

To account for the nucleus's motion in a hydrogen atom, we replace the electron with a particle of mass m' . The atom's energy levels then become

$$E'_n = -\frac{m'e^4}{8 \epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \left(\frac{m'}{m} \right) \frac{E_1}{n^2}$$

Energy levels
corrected for
nuclear motion

$$E_n = -\frac{me^4}{8 \epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \frac{E_1}{n^2}$$

$$-\frac{e^4}{8 \epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \frac{E_1}{mn^2}$$

$$E_1 = -\frac{e^4}{8 \epsilon_0^2 h^2}$$

Due to the nucleus's motion, all hydrogen energy levels shift by the fraction

$$\frac{m'}{m} = \frac{M}{M + m} = 0.99945$$

This represents an increase of 0.055 percent because the energies E_n , being smaller in absolute value, are therefore less negative.

Energy levels
corrected for
nuclear motion

$$E'_n = -\frac{m'e^4}{8\epsilon_0^2 h^2} \left(\frac{1}{n^2} \right) = \left(\frac{m'}{m} \right) \left(\frac{E_1}{n^2} \right)$$

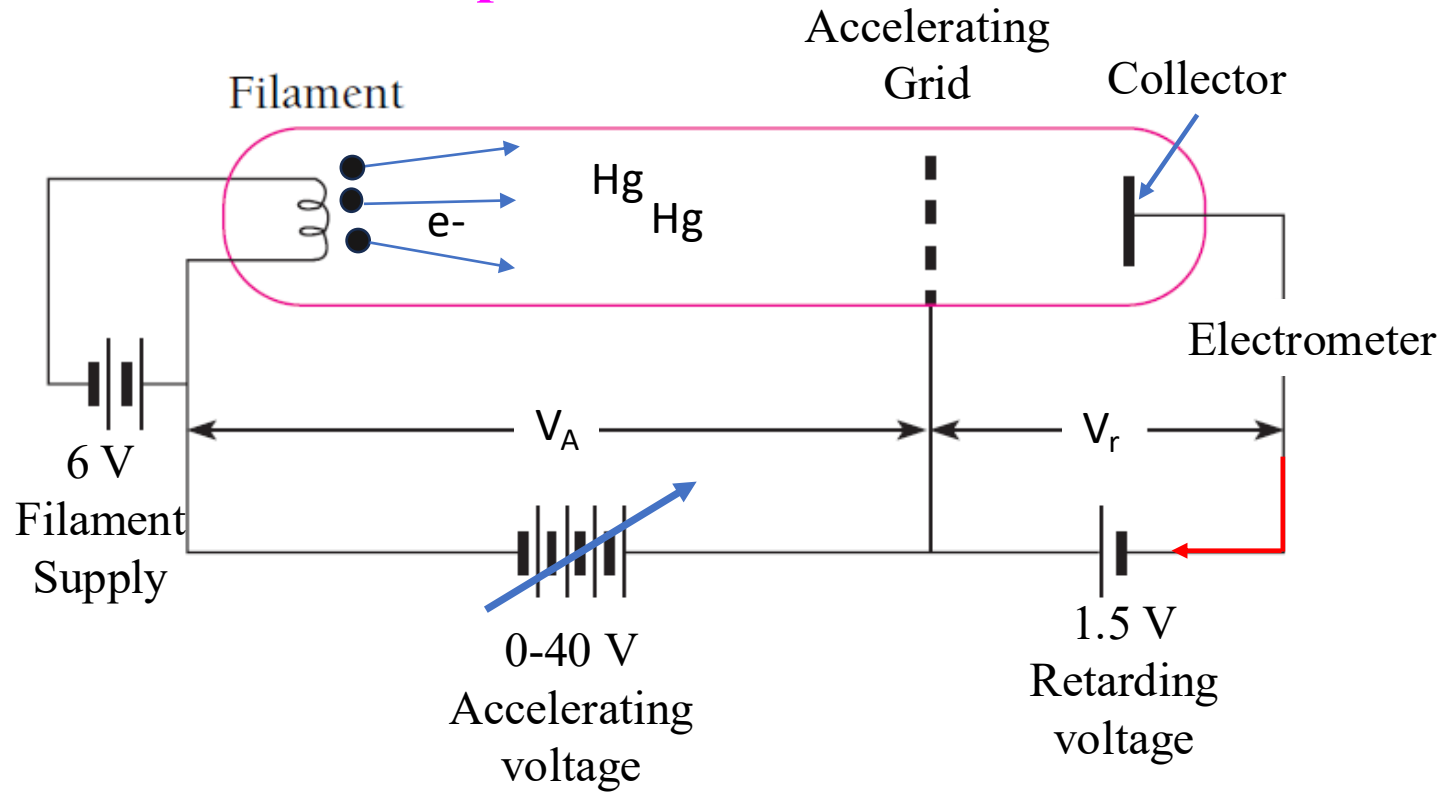
H atom $E = -13.606 \text{ eV}$ $\rightarrow$ -13.598 eV $\leftarrow$ reduced mass

Experiment $E = -13.598 \text{ eV}$

- The energy equation removes the discrepancy between the predicted wavelengths of the spectral lines of hydrogen and the measured ones.
- The value of the **Rydberg constant R** without correcting for nuclear motion is **$1.0973731 \times 10^7 \text{ m}^{-1}$** ;
- The correction lowers **R** to **$1.0967758 \times 10^7 \text{ m}^{-1}$** .

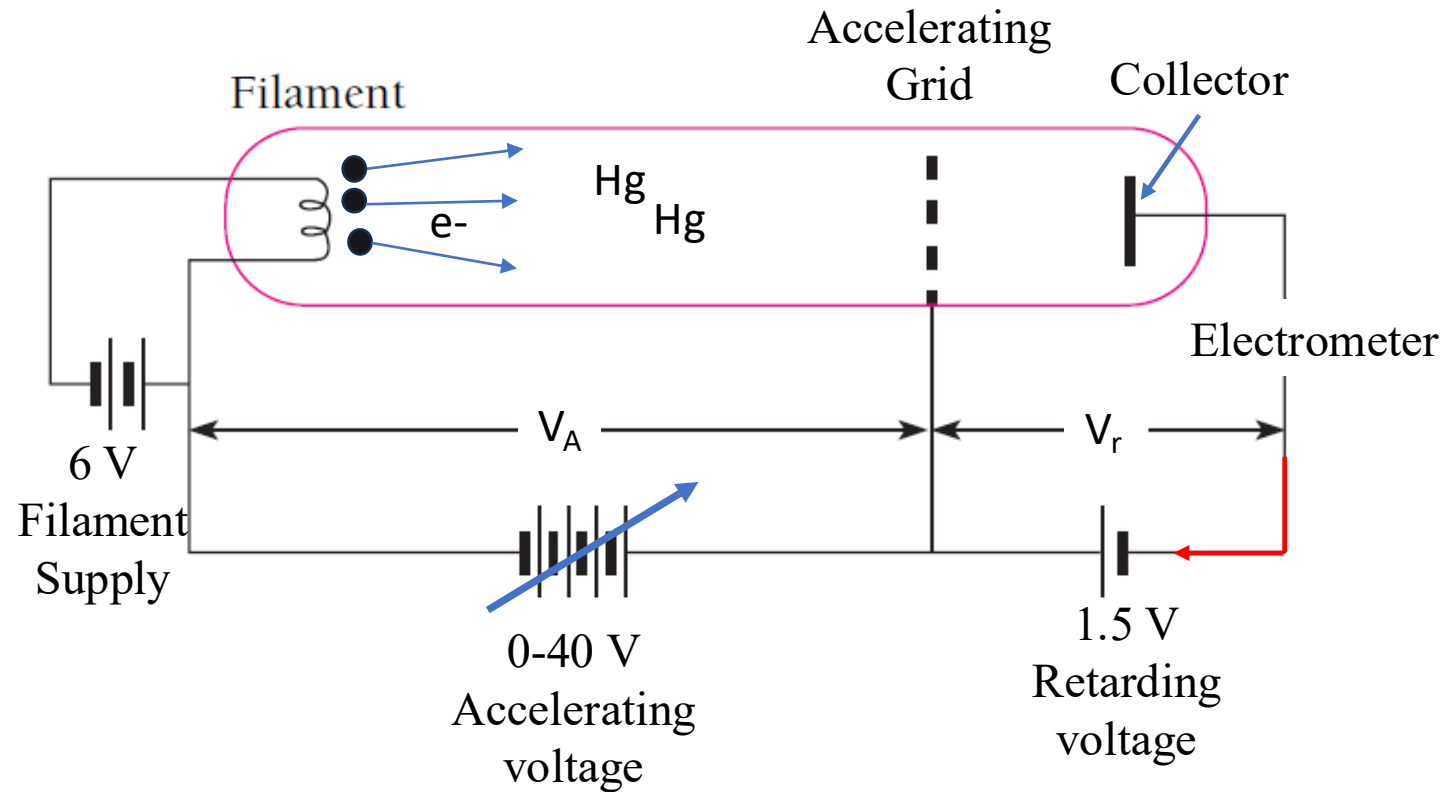
- The notion of reduced mass played an important part in the discovery of **deuterium**,
- Deuterium nucleus contains a neutron as well as a proton.
- Because of the greater nuclear mass, the spectral lines of deuterium are all shifted slightly to wavelengths shorter than the corresponding ones of ordinary hydrogen.
- H α line of deuterium $\rightarrow$ Transition $n = 3 \rightarrow n = 2$ energy level $\rightarrow$ Wavelength 656.1 nm,
- H α line of Hydrogen $\rightarrow$ Transition $n = 3 \rightarrow n = 2$ energy level $\rightarrow$ Wavelength 656.3 nm.
- Identification of deuterium in 1932 by the American chemist Harold Urey.

Franck-Hertz Experiment

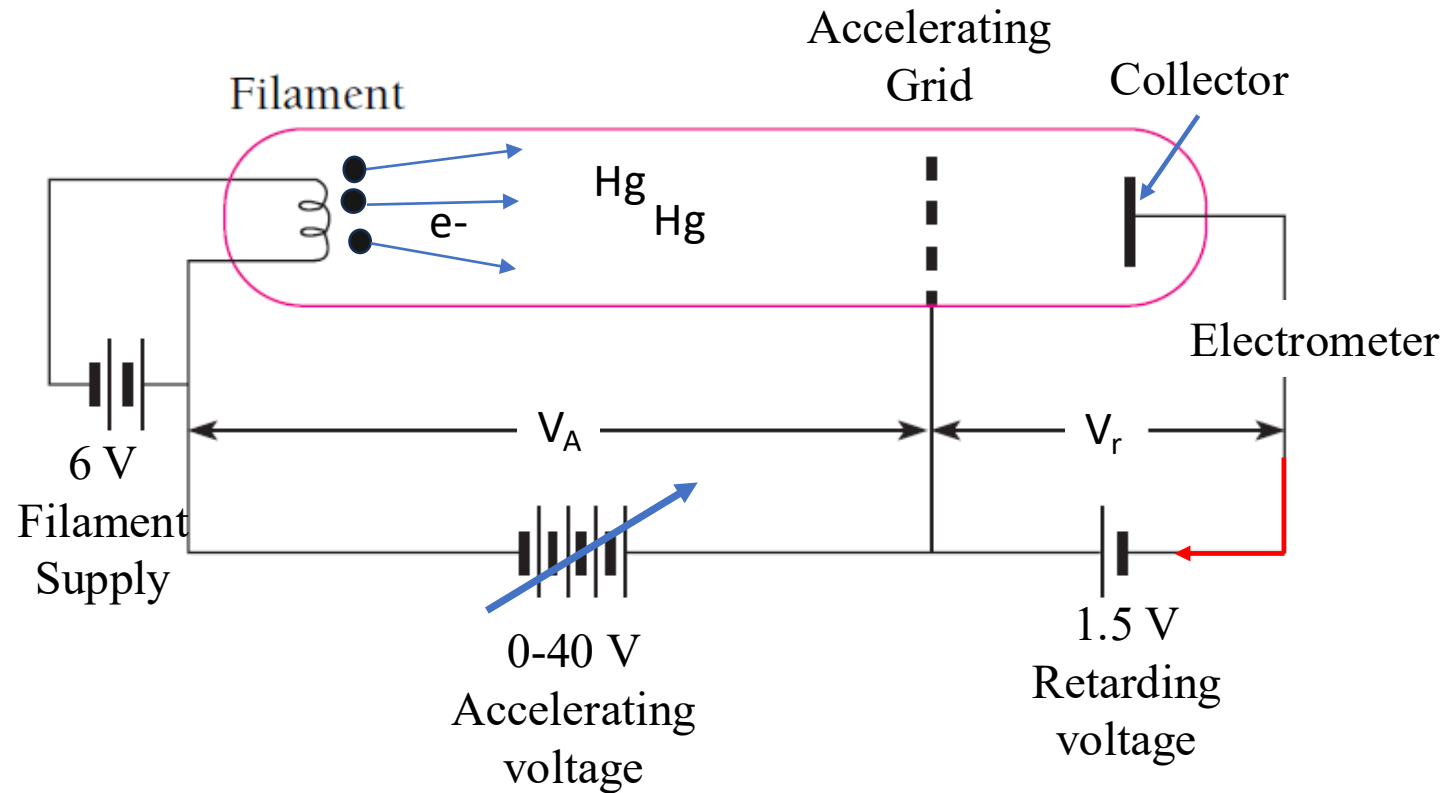


- The Filament ejects electrons in the gas filled tube.
- A variable (positive) accelerating voltage, V_A , is applied to the grid.
- The electron acquire kinetic energy $K=eV_A$

- There is (negative) retarding voltage (V_r) between the grid and collector.
- Only electrons having enough energy will overcome this potential and reach the collector and contribute to current measured with an Electrometer.

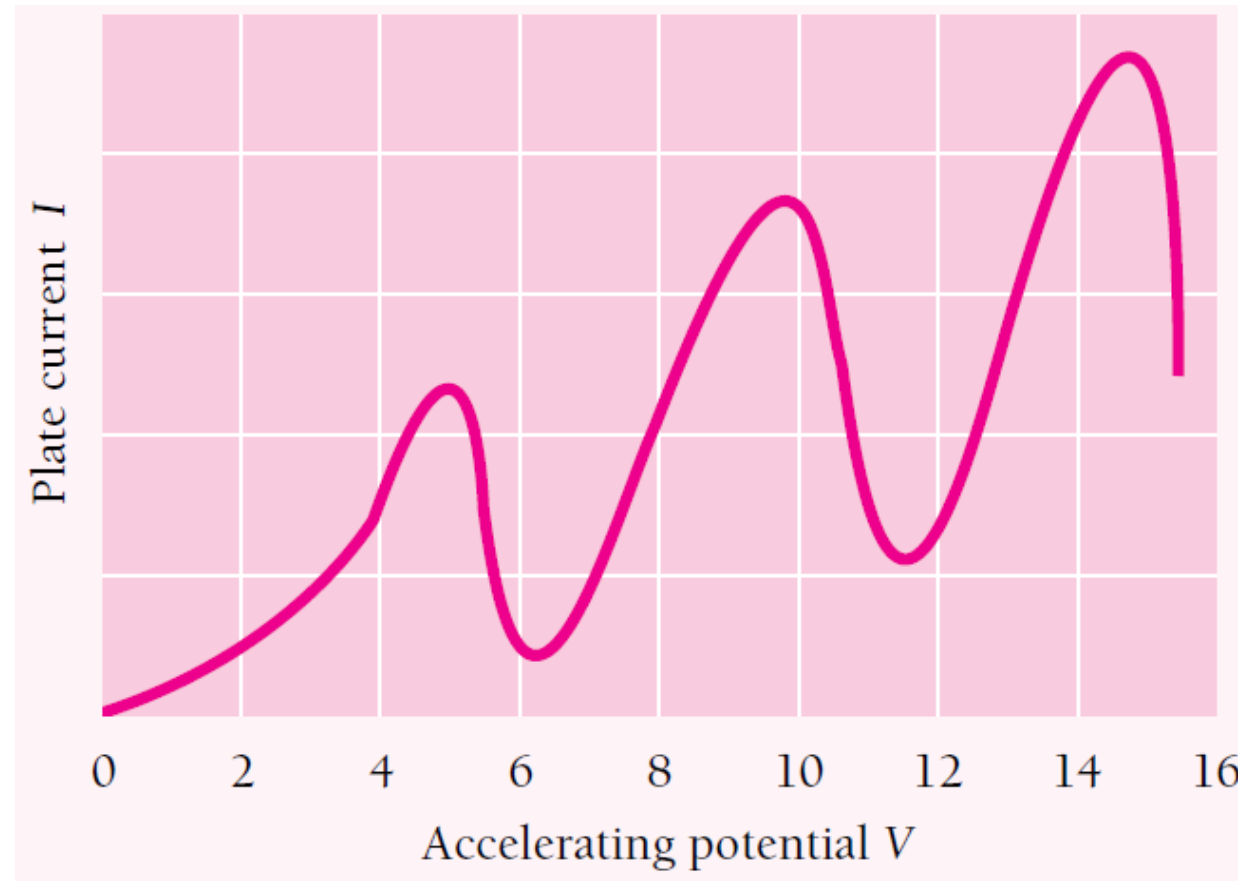


- If $V_r > V_A \rightarrow$ no electron can reach the collector, so no current would be measured.
- If $V_r < V_A \rightarrow$ if the tube is highly evacuated, most of the electrons would reach the collector and have energy $|e|(V_A - V_r)$

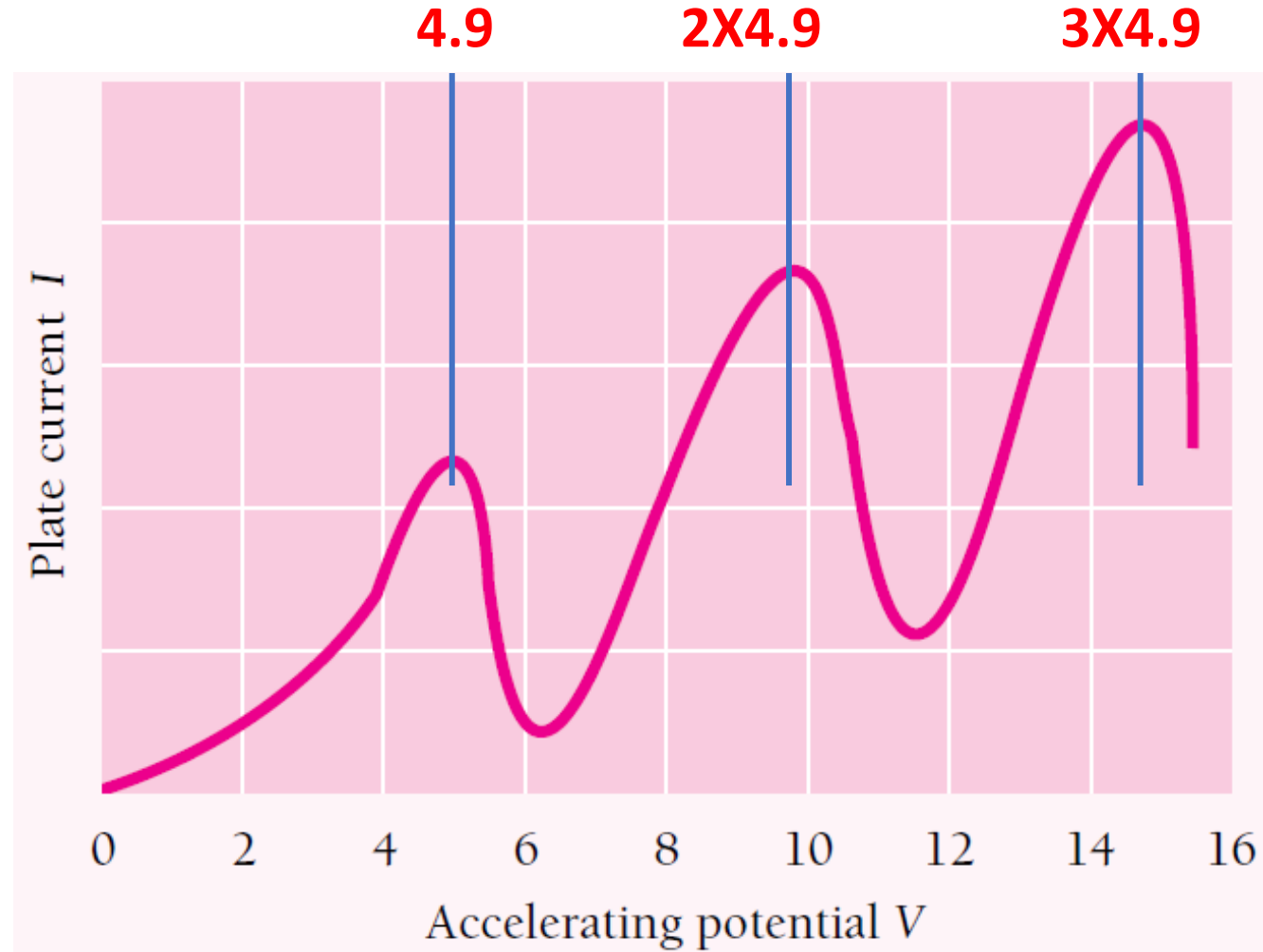


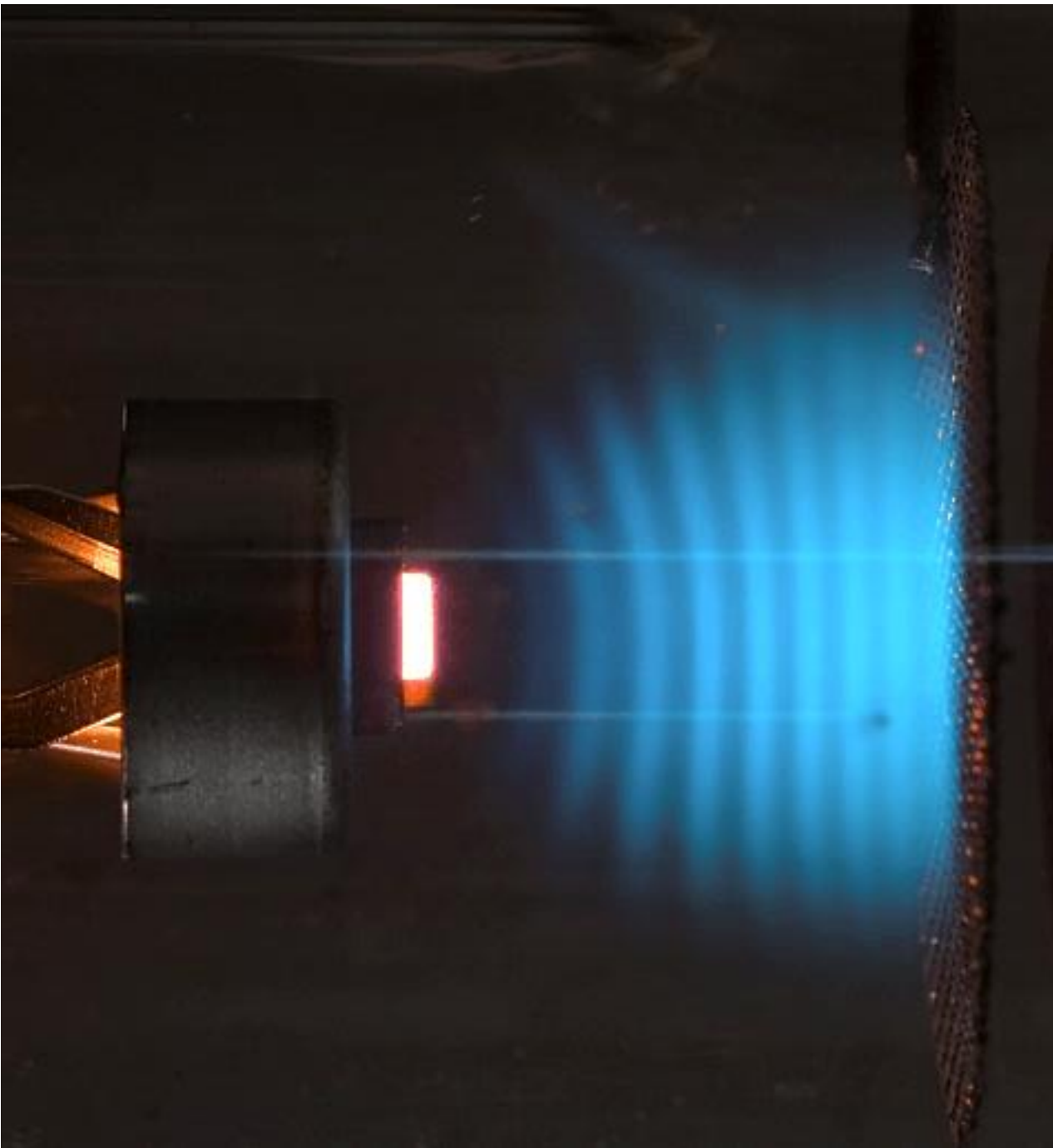
- If the tube contains some gas, the electrons can lose energy via collisions with the gas atoms.
- Such collisions are inelastic, i.e. electrons lose energy, which is transferred to internal energy of atoms in the gas.
- Thus, even in the case when $V_r < V_A$, it is possible that **the electrons would not be able to reach collector**, and contribute to the current.

- Franck and Hertz observed the collector current as a function of $V_A (>V_r)$ when tube was filled with various gases (result for mercury gas is shown here)
- At first, the current increased as was expected for a typical vacuum tubes, **but as $\sim 4.9\text{V}$ current suddenly dropped.**
- Then, the increase resumed until 9.8 V , and so on.



- The current drops because fewer electrons reach the collector.
- This occurs only if the electrons undergo elastic collisions.
- Thus, when $V_A = 4.9 \times n$ Volts ($n=1, 2, \dots$) the electrons undergo inelastic collisions with Hg atoms.
- In inelastic collision kinetic energy of electrons is transferred to the internal energy of Hg atoms. Hg atoms absorb the energy of electrons.





Concludes chapter 4.

For any science-related doubts,
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